# Basal Sliding

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Caltech Ge 193, Spring 2026

## Topics Covered

- The three mechanisms of glacier motion: internal deformation, sliding over the bed, and bed deformation
- Hard-bed sliding: regelation and enhanced creep (Weertman's theory)
- The role of water pressure: effective pressure, cavities, and Iken's bound
- Deformable beds: till as a Mohr–Coulomb plastic material
- Phenomenological sliding relations used in glacier models
- Local vs. global control of basal velocity

**Primary reference:** Cuffey & Paterson, *Physics of Glaciers*, 4th ed., Chapter 7.

## 1 Introduction

Glaciers move by three mechanisms: plastic deformation of the ice, sliding of ice over the bed, and deformation of the bed itself. These are not mutually exclusive—sliding and bed deformation often occur together. Collectively, the motion arising from sliding and bed deformation is called **basal slip**. Basal slip accounts for nearly all of the flow of the ice streams that drain ice from Antarctica, and it dominates the motion of most fast-flowing glaciers worldwide. Understanding basal slip is therefore essential for predicting the response of glaciers and ice sheets to climate forcing.

The difficulty is that the glacier bed is hidden. Direct observations are rare and come mostly from boreholes, subglacial tunnels, and cavities—none of which are necessarily representative of typical conditions. As a result, complex mathematical theories of sliding have been developed, but they have limited predictive value. They do, however, contain important physical insights that guide the formulation of phenomenological relations used in glacier models.

Beds are classified as **hard** (rigid, impermeable bedrock) or **soft** (deformable, permeable sediment, called till). The physics differs fundamentally between the two cases. On a hard bed, the resistance to sliding depends on the roughness of the bedrock and the processes of regelation and ice creep. On a soft bed, the resistance depends on the strength of the sediment, which is controlled by the effective pressure. In both cases, water pressure plays a central role.

**Plan for this lecture.** We begin with the distinction between local and global control of basal velocity (§2). We then develop the classical theory of hard-bed sliding (§3), including Weertman's analysis and the critical role of water pressure and cavities. Next, we discuss deformable beds and the Mohr–Coulomb description of till (§4). Finally, we discuss the phenomenological sliding relations used in glacier dynamics models (§5).

## 2 Local vs. Global Control of Basal Velocity

To understand what determines the slip rate $u_b$, two end-member situations must be distinguished.

### 2.1 Local Control

In the first end member, the bed provides all or most of the resistance to the glacier's flow. The basal shear stress $\tau_b$ is determined by the gravitational driving stress, and the local slip relation $u_b(\tau_b)$ determines the slip rate. If conditions at the bed change—for example, if water pressure increases and the bed becomes more slippery—then $u_b$ increases. This situation is called **local control** (or dynamic control).

### 2.2 Global Control

In the second end member, the bed provides only fractional resistance and deformation of the ice in a large surrounding region is important, including shearing at the lateral margins, stretching along flow, and flow over resistant patches elsewhere on the bed. This is called **global control** (or nonlocal control, or kinematic control). A floating ice shelf is the extreme limit—the bed provides negligible resistance. West Antarctic ice streams, with basal shear stresses of only a few kPa, lie close to this limit.

Most glaciers lie somewhere between these end members. The key implication is that on a weak or well-lubricated bed, the sliding rate cannot generally be predicted from local bed properties alone. This is why no single "sliding law" relating $u_b$ to $\tau_b$ can be universally valid and is also why can't trade space for time when it comes to inferring sliding laws.

## 3 Hard-Bed Sliding

### 3.1 Weertman's Theory: Regelation and Enhanced Creep

Modern study of hard-bed sliding dates from the analysis of [19]. The problem is to explain how ice at the pressure melting point moves past bumps on a rigid, impermeable bedrock. Two mechanisms operate:

1. **Regelation.** The pressure is elevated on the upstream side of a bump and reduced on the downstream side. Because the melting point of ice decreases with increasing pressure, ice melts on the upstream side and the meltwater flows around to the downstream side, where it refreezes. The latent heat released by refreezing is conducted back through the bump to sustain the melting. This mechanism is most effective for *small* bumps, because the heat must be conducted through the obstacle.

2. **Enhanced creep.** A bump in the bed produces a stress concentration that drives viscous deformation of the ice around the bump. This mechanism is most effective for *large* bumps, because the volume of ice affected—and hence the distance over which enhanced straining occurs—scales with the bump size.

Weertman idealized the bed as an inclined plane studded with cubical bumps of side length $a$, spaced a distance $\lambda$ apart (Figure 1). Define the **roughness** $R = a/\lambda$. We now derive the sliding velocity due to each mechanism.

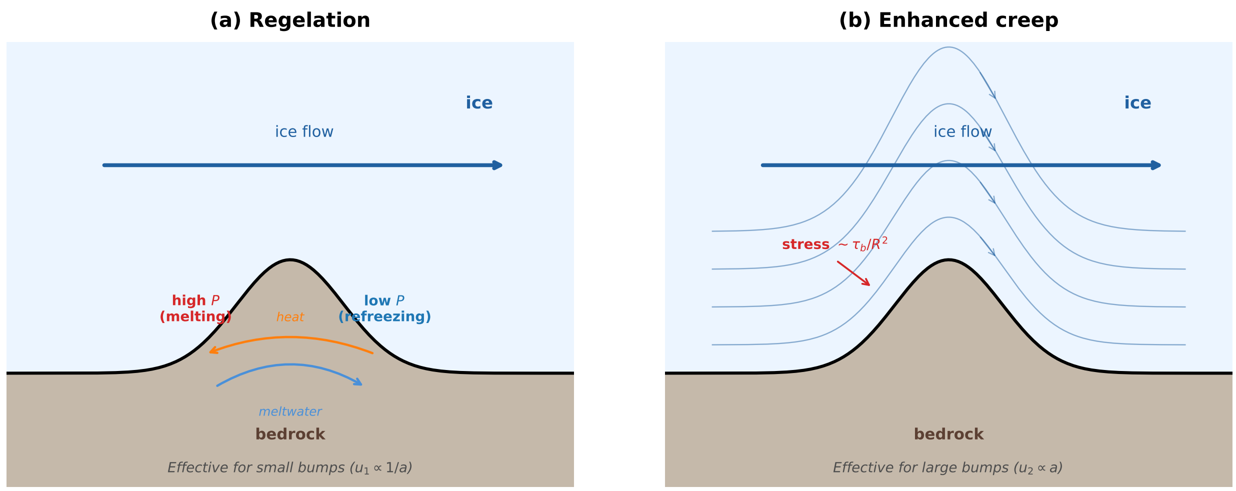


Figure 1: The two mechanisms of hard-bed sliding. **(a)** Regelation: elevated pressure on the upstream side of a bump lowers the melting point, causing ice to melt; meltwater flows around the bump and refreezes on the low-pressure downstream side; latent heat released by refreezing is conducted back through the bump. Effective for small bumps ($u_1 \propto 1/a$). **(b)** Enhanced creep: the stress anomaly $\sim \tau_b/R^2$ produced by the bump drives viscous deformation of ice over and around the obstacle; flowlines (blue) are deflected around the bump. Effective for large bumps ($u_2 \propto a$).

#### 3.1.1 Step 1: Force balance and the stress anomaly

Consider one bump with upstream and downstream faces each of area $a^2$. Each bump is isolated within a broader patch of bed of area $\lambda^2$. The applied driving stress $\tau_d$ acts uniformly over this patch, so the total horizontal force on the bump is

$$F = \tau_d \, \lambda^2. \tag{1}$$

We assume a thin layer of water separating the ice and the bed (i.e., zero shear stresses at the ice-bed interface), meaning this force must be supported by a pressure difference between the upstream and downstream faces. Let $\Delta p$ be the pressure anomaly so that the upstream face experiences a pressure $\bar{p} + \Delta p$ and the downstream face a pressure $\bar{p} - \Delta p$, where $\bar{p}$ is the ambient pressure. The net horizontal force from this pressure difference is $2\,\Delta p\, a^2$. Equating this to $F$:

$$2\,\Delta p\, a^2 = \tau_d\, \lambda^2 \qquad \Longrightarrow \qquad \Delta p = \frac{\tau_d}{2}\left(\frac{\lambda}{a}\right)^2 = \frac{\tau_d}{2\,R^2}. \tag{2}$$

Because $R \ll 1$ in practice (typical values $R \sim 0.01$–$0.1$), the stress anomaly $\Delta p$ is much larger than the applied shear stress $\tau_b$. For example, $R = 0.05$ gives $\Delta p = 200\,\tau_b$. This stress concentration is what makes both regelation and enhanced creep effective.

#### 3.1.2 Step 2: Regelation velocity

The Clausius–Clapeyron relation states that the melting point of ice decreases with pressure:

$$T_m = T_0 - \mathcal{B}\, p, \tag{3}$$

where $T_0 = 273.15$ K is the melting point at zero pressure and $\mathcal{B} \approx 7.4 \times 10^{-5}$ K (kPa)$^{-1}$ is the Clausius–Clapeyron slope. The pressure difference $2\,\Delta p$ between the upstream and downstream faces therefore produces a difference in the pressure melting point:

$$\Delta T = \mathcal{B} \cdot 2\,\Delta p = \frac{\mathcal{B}\,\tau_d}{R^2}. \tag{4}$$

The ice is everywhere at the pressure melting point. On the upstream face, the elevated pressure depresses the melting point below the temperature of the arriving ice, so ice melts. On the downstream face, the reduced pressure raises the melting point above the temperature of the arriving meltwater, so it refreezes. Because the ice is at the melting point on both faces, $\Delta T$ is also the temperature difference across the rock bump: the upstream face of the rock is in contact with ice at $T_m(\bar{p} + \Delta p)$ and the downstream face with ice at $T_m(\bar{p} - \Delta p)$. This temperature difference drives heat conduction through the rock from the downstream side (where refreezing releases latent heat) to the upstream side (where melting absorbs it).

Treating the bump as a slab of bedrock with thermal conductivity $k_r$, the heat flux per unit area driven by $\Delta T$ across the bump width $a$ is

$$q = k_r\,\frac{\Delta T}{a} = \frac{k_r\,\mathcal{B}\,\tau_d}{a\,R^2}. \tag{5}$$

This heat flux melts a thickness of ice per unit time on the upstream face. The mass of ice melted per unit area per unit time is $q/L$, where $L$ is the specific latent heat of fusion. The regelation velocity—the rate at which ice moves past the bump by the melt–refreeze cycle—is the volume melted per unit time divided by the face area:

$$\boxed{u_1 = \frac{k_r\,\mathcal{B}}{\rho\,L\,a}\,\frac{\tau_d}{R^2}.} \tag{6}$$

The critical feature is that $u_1 \propto 1/a$: the regelation velocity is inversely proportional to the bump size. Heat must be conducted through the entire width of the bump, so smaller bumps offer less resistance to regelation. For large bumps, the conduction path is long and regelation becomes negligibly slow.

Numerically, for $k_r = 3\ \mathrm{W\,m^{-1}\,K^{-1}}$ (granite), $\rho = 917\ \mathrm{kg\,m^{-3}}$, $L = 3.34\times10^5\ \mathrm{J\,kg^{-1}}$, $\tau_d = 100$ kPa, $R = 0.05$, and $a = 1$ mm:

$$u_1 = \frac{3 \times 7.4 \times 10^{-8}}{917 \times 3.34 \times 10^5 \times 10^{-3}} \times \frac{10^5}{0.05^2} \approx 29\ \mu\mathrm{m\,s^{-1}} \approx 910\ \mathrm{m\,yr^{-1}}.$$

For $a = 1$ m, the same calculation gives $u_1 \approx 0.9\ \mathrm{m\,yr^{-1}}$—negligible. Regelation is rapid, and thus not rate-controlling, for millimeter-scale bumps and negligibly slow for meter-scale bumps.

#### 3.1.3 Step 3: Enhanced creep velocity

The stress anomaly $\Delta p = \tau_d/(2R^2)$ also drives viscous deformation of the ice around the bump. The ice must flow over and around the obstacle, and in the classical example, the rate at which it does so is determined by Glen's flow law. The effective strain rate in the vicinity of the bump is

$$\dot{\varepsilon} \sim A\,(\Delta p)^n = A\left(\frac{\tau_d}{2\,R^2}\right)^n, \tag{7}$$

where $A$ is the creep parameter and $n$ is the creep exponent. The ice must deform over a distance comparable to the bump size $a$ to flow past the obstacle, so the enhanced creep velocity scales as

$$\boxed{u_2 \sim a\,A\left(\frac{\tau_d}{2\,R^2}\right)^n.} \tag{8}$$

(The exact numerical prefactor depends on the bump geometry and the solution of the full Stokes problem; Weertman obtained $u_2 = 2 \cdot 3^{-(n+1)/2}\, a\, A\, [\tau_d/(2R^2)]^n$.)

The essential feature is that $u_2 \propto a$: the creep velocity is directly proportional to the bump size. Larger bumps produce the same stress anomaly (for fixed $R$), but the ice must deform over a greater distance, so the velocity contribution is larger. For small bumps, the deformation distance is short and enhanced creep is negligible.

### 3.2 The Controlling Obstacle Size

Regelation gives $u_1 \propto 1/a$ and enhanced creep gives $u_2 \propto a$. For any bump size, the total velocity past that bump is $u = u_1 + u_2$. As a function of $a$, this sum has a minimum at an intermediate bump size $a_c$ called the **controlling obstacle size** (Figure 2). Bumps of this size provide the greatest resistance to sliding, because both mechanisms are relatively slow. Bumps smaller than $a_c$ are easily passed by regelation; bumps larger than $a_c$ are easily passed by enhanced creep. The controlling obstacles determine the overall sliding velocity.

The controlling obstacle size is found by setting $u_1 = u_2$:

$$\frac{k_r\, \mathcal{B}\, \tau_d}{\rho\, L\, a_c\, R^2} = a_c\, A \left(\frac{\tau_d}{2\, R^2}\right)^n. \tag{9}$$

Solving for $a_c^2$:

$$a_c^2 = \frac{k_r\, \mathcal{B}}{\rho\, L\, A}\, \frac{\tau_d}{R^2} \left(\frac{2\, R^2}{\tau_d}\right)^n = \frac{2^n\, k_r\, \mathcal{B}}{\rho\, L\, A}\, \tau_d^{1-n}\, R^{2n-2}, \tag{10}$$

so

$$a_c = \left(\frac{2^n\, k_r\, \mathcal{B}}{\rho\, L\, A}\right)^{1/2} \tau_d^{(1-n)/2}\, R^{n-1}. \tag{11}$$

For $n = 3$ and typical parameter values, $a_c$ is in the range 10–100 mm.

### 3.3 Weertman's Sliding Law

The total sliding velocity at the controlling obstacle size is $u_b = u_1(a_c) + u_2(a_c) = 2\, u_1(a_c)$, since $u_1 = u_2$ at $a = a_c$. Substituting equation (11) into equation (6):

$$\begin{aligned} u_b &= 2\, \frac{k_r\, \mathcal{B}\, \tau_d}{\rho\, L\, a_c\, R^2} = 2\, \frac{k_r\, \mathcal{B}\, \tau_d}{\rho\, L\, R^2} \left(\frac{\rho\, L\, A}{2^n\, k_r\, \mathcal{B}}\right)^{1/2} \tau_d^{(n-1)/2}\, R^{-(n-1)} \\ &= 2 \left(\frac{k_r\, \mathcal{B}\, A}{2^n\, \rho\, L}\right)^{1/2} \tau_d^{(n+1)/2}\, R^{-(n+1)}. \end{aligned} \tag{12}$$

This can be written compactly as

$$\boxed{u_b = \text{const} \cdot \left[\frac{\tau_d^{1/2}}{R}\right]^{n+1},} \tag{13}$$

where the constant absorbs all of the material parameters ($k_r$, $\mathcal{B}$, $A$, $\rho$, $L$). For $n = 3$, the sliding velocity varies as the **square** of the driving stress ($u_b \propto \tau_d^2$) and inversely as the **fourth power** of the roughness ($u_b \propto R^{-4}$).

The strong dependence on roughness—much stronger than on stress—is a fundamental prediction of the theory. It means that bed geometry is more important than applied stress in determining the sliding rate.

### 3.4 Nye–Kamb Theory and Extensions

[13, 14] and [11] improved Weertman's analysis by treating the bed as a continuous rough surface described by a Fourier spectrum, rather than as isolated cubical bumps. They assumed Newtonian ice ($n = 1$) and solved the full Stokes equations for flow over a wavy bed with regelation at the ice–bed interface. Their result has the same form as Weertman's: $u_b \propto \tau_d / R^2$ (equation (13) with $n = 1$), with the roughness now defined as a spectral property of the bed topography.

[3, 4] extended the analysis to nonlinear ice and a sinusoidal bed of wavelength $\lambda$ and amplitude $a$:

$$u_b = c \, \frac{\lambda}{R^{n+1}} \, A \, \tau_d^n, \tag{14}$$

where $c$ is a coefficient that depends on $n$ and the ratio $\lambda/H$ (ice thickness).

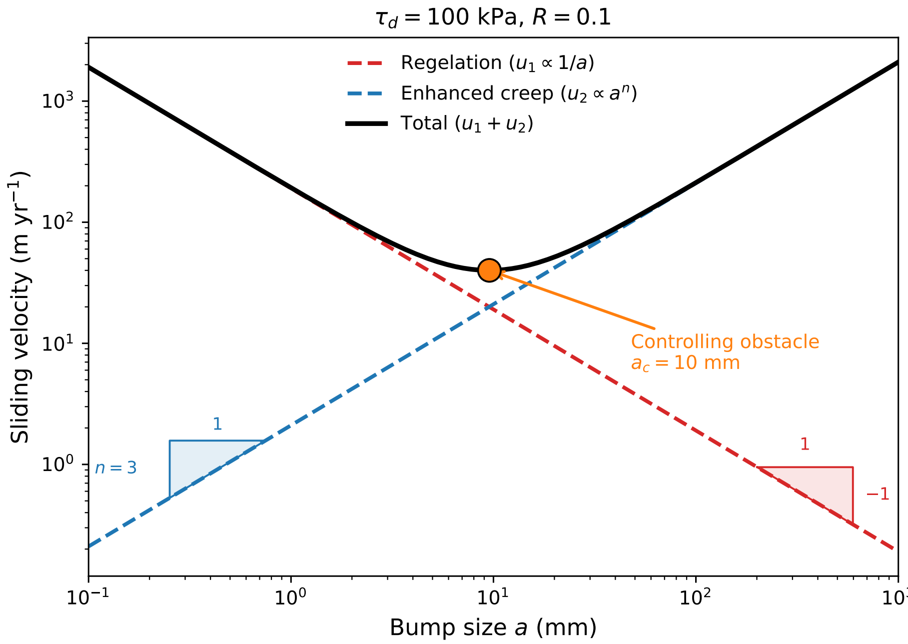


Figure 2: Sliding velocity contributions from regelation ($u_1 \propto 1/a$) and enhanced creep ($u_2 \propto a$, at fixed roughness $R$) as functions of bump size $a$ for $\tau_d = 100$ kPa and $R = 0.1$. The controlling obstacle size $a_c$ (orange circle) is the bump size at which the total velocity $u_1 + u_2$ is minimized; bumps of this size provide the greatest resistance to sliding.

**Key point: limitations of hard-bed sliding theory**

Weertman-type theories predict sliding velocities that are broadly consistent with observations on slow-sliding mountain glaciers (drag factors $\psi = \tau_b/u_b$ of order 1–20 kPa (m yr$^{-1}$)$^{-1}$; Table 1). They fail, however, to explain fast sliding ($u_b > 20$ m yr$^{-1}$), because they omit a critical ingredient: water-filled cavities in the lee of bumps. Fast sliding on a hard bed requires cavities, and cavities require water pressures that approach the ice-overburden pressure.

Table 1: Apparent drag factors $\psi = \tau_b/u_b$ for selected glaciers, from [2]. Glaciers with small $\psi$ (slippery beds) tend to be fast-flowing, marine-terminating, or underlain by deformable sediment.

| **Glacier** | **Terminus type** | $\tau_b$ (kPa) | $u_b$ (m yr$^{-1}$) | $\psi$ (kPa (m yr$^{-1}$)$^{-1}$) |
|---|---|---|---|---|
| Ice shelves | marine | $\sim$0 | 100–1000 | 0 |
| Whillans Ice Stream | marine | 3 | 400 | 0.008 |
| Columbia Glacier | marine | 100–150 | 1000–2000 | 0.05–0.2 |
| Storglaciären | land | 40 | 30 | 1 |
| Engabreen | land | 100 | 44 | 2 |
| Haut Gl. d'Arolla | land | 120 | 6 | 20 |

## 3.5 Effective Pressure and Cavity Formation

The analyses of §3.1 assume that all basal water resides in a thin film between ice and rock. In reality, water also fills cavities in the lee of bedrock bumps. Cavities reduce the area of contact between ice and bed, effectively reducing the roughness and increasing the sliding velocity. The formation and evolution of cavities is governed by the **effective pressure**:

$$\boxed{N = P_i - P_w = \rho_i\, g\, H - P_w,} \tag{15}$$

where $P_i = \rho_i g H$ is the ice-overburden pressure and $P_w$ is the subglacial water pressure. When $N \to 0$, the glacier approaches flotation and cavities can grow without bound.

### 3.5.1 The Separation Pressure

For a sinusoidal bed with wavelength $\lambda$ and amplitude $a$, the normal stress exerted by the ice on the bed fluctuates about the overburden value $P_i$. The minimum normal stress—and hence the water pressure at which ice first separates from the bed—is the **separation pressure**:

$$P_s = P_i - \frac{\lambda\, \tau_b}{a\, \pi}. \tag{16}$$

Cavities begin to form when $P_w$ reaches $P_s$. The rougher the bed (larger $a/\lambda$), the higher the water pressure required to initiate cavitation. Figure 3 illustrates the progressive stages of cavity growth.

### 3.5.2 Iken's Bound on Basal Drag

[8] derived an upper limit on the basal shear stress that a hard bed can support. For a bed with bumps whose steepest upstream face makes an angle $\beta$ with the mean bed, the critical effective pressure at which sliding becomes unstable is

$$\boxed{\tau_b = N_c\, \tan\beta,} \tag{17}$$

where $N_c = P_i - P_c$ is the critical effective pressure. For a sinusoidal bed, $\tan\beta = 2\pi a/\lambda$, so

$$P_c = P_i - \frac{\lambda\, \tau_b}{2\pi\, a}. \tag{18}$$

This result establishes a fundamental constraint: the drag that a hard bed can exert on the overlying ice is bounded above by a value proportional to the effective pressure and the maximum bed slope. Once water pressure exceeds $P_c$, a net upward force acts on the ice and sliding becomes unstable—formally analogous to the Coulomb friction criterion for landslides. [15] proved that this bound holds for arbitrary (non-sinusoidal) bed geometries, provided the bed has no vertical faces.

### 3.5.3 Schoof's Cavitation Theory

[15] placed these ideas on a rigorous footing. He considered the Nye–Kamb sliding problem for a Newtonian fluid flowing over a periodic hard bed with cavitation, extending Fowler's (1986) earlier model to allow any finite number of cavities per bed period and arbitrary (non-sinusoidal) bed geometries. Three key results emerged:

1. **Iken's bound holds for arbitrary beds.** For any bed whose slope is bounded by $\max(h') = m_{\max}$, the drag satisfies $\tau_b \leq N\, m_{\max}$, where $m_{\max}$ plays the role of $\tan\beta$. The only geometric quantity that matters for the upper bound is the steepest upstream face of the bed.

2. **The sliding law is a relation $\tau_b/N = f(u_b/N^n)$.** The drag-to-effective-pressure ratio depends on slip velocity only through the combination $\Lambda = u_b/N^n$. This follows from the scaling of the Stokes equations used in the boundary-layer analysis: stress scales as $N$, velocity perturbations scale as $u_b$, and the problem is parameterized by their ratio.

3. **The drag reaches a maximum and then decreases.** As $\Lambda$ increases from zero, $\tau_b/N$ first rises (the Weertman regime, where drag increases with velocity), reaches a global maximum (typically less than $N\, m_{\max}$), and then decreases as contact areas migrate toward the tops of bumps and the effective bed roughness is reduced. The maximum drag is controlled by the bed obstacles with the steepest slopes.

The existence of a maximum in the $\tau_b$–$u_b$ relation is a robust prediction of all cavity-based sliding theories. It means that above a certain sliding speed, faster sliding produces *less* drag—a velocity-weakening regime that has important consequences for glacier stability and surge dynamics. [15] proposed a simple analytical form that captures these features:

$$\frac{\tau_b}{N} = C\left(\frac{\Lambda}{\Lambda + \Lambda_0}\right)^{1/n}, \quad \Lambda = \frac{u_b}{N^n}, \tag{19}$$

where $C < m_{\max}$ is a constant related to the bed geometry, $\Lambda_0 \sim \lambda_{\max} A/m_{\max}$ is a reference velocity scale set by the dominant obstacle wavelength $\lambda_{\max}$ and slope $m_{\max}$, and $n$ is the exponent in Glen's flow law. At low velocities ($\Lambda \ll \Lambda_0$), equation (19) gives $\tau_b \propto u_b^{1/n}$ (rate-strengthening). At high velocities ($\Lambda \gg \Lambda_0$), $\tau_b \to C\,N$ (rate-independent). Figure 4 shows this relationship for three values of $N$.

## 3.6 Observations: Velocity and Water Pressure

The theoretical expectation that sliding velocity should depend on effective pressure is well supported by observations. Two classic datasets illustrate this:

**Findelengletscher** [9]. Surface velocity and borehole water levels were measured at four lines of stakes over five weeks early in the melt season. Velocity and water pressure correlate on timescales of days to weeks. When plotted against effective pressure $N$, velocity varies approximately as $N^{-1}$.

**Lauteraargletscher** [16]. Diurnal cycles of velocity and water pressure show a clear correlation, with a notable asymmetry: sliding is faster when water pressure is rising (and $N$ is decreasing) than when it is falling (and $N$ is increasing) at the same value of $N$. This hysteresis likely reflects the finite time required for cavities to adjust to changing water pressure—transient cavity growth temporarily contributes to sliding.

**Bench Glacier** [6, 7]. Two speed-up events were observed in which neither was accompanied by a reduction in effective pressure. Instead, bed separation increased (indicating cavity growth), but without a decrease in $N$. This demonstrates that sliding rate can increase without changes in effective pressure, through the redistribution or increased volume of water at the bed.

**Key point: water pressure is necessary but not sufficient**

Water at the glacier bed increases the sliding rate through at least four mechanisms:

1. Low effective pressures allow large cavities, reducing the area of ice–bed contact.
2. Water can spread across the bed or fill new cavities even without decreasing effective pressure.
3. Reduced effective pressures decrease frictional drag from rock particles in the basal ice.
4. Transient cavity growth displaces ice downstream, temporarily increasing the sliding rate.

Both the effective pressure $N$ and the volume of water per unit area of bed must be considered as independent variables in any complete sliding relation.

### 3.7 Hard-Bed Sliding: Summary

The essential features of hard-bed sliding are:

1. Ice moves past bedrock bumps by a combination of regelation (dominant for small bumps) and enhanced creep (dominant for large bumps). The controlling obstacle size—for which both processes are equally effective—is in the range 10–100 mm.
2. Rapid sliding ($u_b \gtrsim 20\ \mathrm{m\,yr^{-1}}$) on a hard bed requires water-filled cavities in the lee of bumps. Without cavities, theoretical sliding speeds are much lower than observed.
3. The drag that a hard bed can support is bounded above by $\tau_b \leq N \tan\beta$ (Iken's bound). As water pressure approaches the overburden, the bed becomes unable to resist the flow and the velocity is controlled by global stresses—lateral drag, longitudinal stresses, and flow over sticky spots elsewhere.
4. Friction between rock particles in the basal ice and the bedrock can be a significant component of the total drag, potentially equaling or exceeding the drag from regelation and creep around bedrock bumps. No existing theory provides a complete quantitative treatment of hard-bed sliding that includes regelation, creep, cavitation, rock–rock friction, and water flow simultaneously.

## 4 Deformable Beds

Many glacier beds are not rigid bedrock but consist of unconsolidated sediment—till in glaciological usage. Deformable beds underlie vast areas of the Pleistocene ice sheets in North America and Europe and are common beneath modern glaciers. The West Antarctic ice streams, which account for the majority of ice discharge from the West Antarctic Ice Sheet, rest on layers of weak, water-saturated till only a few meters thick.

## 4.1 Key Observations

Table 2 summarizes borehole observations from glaciers known to have deforming beds. Both soft-bed sliding (at the ice–till interface) and distributed deformation within the till contribute to the total slip rate. On average, sliding at the interface accounts for the majority of the motion.

Table 2: Observations of subglacial deformation from borehole studies. After [2], Table 7.4.

| **Glacier** | $u_b$ ($\mathrm{m\,yr^{-1}}$) | **Character** | **Bed stress** (kPa) | $N$ (kPa) |
|---|---|---|---|---|
| Whillans | 400 | >80% sliding | <6 | −20 to −150 |
| Bindschadler | 360 | <20% sliding | <10 | 20–40 |
| Columbia | ∼2000 | | ∼100 | |
| Storglaciären | 15–35 | sliding dominates | 40 | 110 |
| Breidamerkur | ∼100 | 10–80% sliding | 80 | 0–200 |

## 4.2 Till as a Plastic Material

Laboratory experiments on subglacial tills consistently show that, after an initial adjustment period, a shearing till attains a yield stress $\tau_*$ that is essentially independent of the deformation rate. This is the behavior of a **perfectly plastic material** at a **critical state**. The yield stress obeys the Mohr–Coulomb criterion [10]:

$$\boxed{\tau_* = c_o + f\,N \approx f\,N,} \tag{20}$$

where $c_o$ is the apparent cohesion (small for deforming tills), $f = \tan\varphi$ is the friction factor, and $N = \sigma_n - P_w$ is the effective normal stress on the failure plane. Measured values of $\varphi$ range from about 10° to 40°, giving $f$ in the range 0.17–0.84. The cohesion is typically small enough to neglect.

The critical implication of equation (20) is that **the strength of till is controlled by water pressure**. Figure 5 and Table 3 show measured values for several subglacial tills. At effective stresses of order 100 kPa, a till with $\varphi = 30°$ has a yield stress of about 60 kPa. But if water pressure rises to within a few percent of overburden ($N \to 0$), the till strength drops to near zero and the bed becomes extremely weak.

Table 3: Measured strength values for subglacial tills. After [2], Table 7.5. Values of $\tau_*$ at $N$ = 50 kPa are reported where $\varphi$ and $c_o$ were measured.

| **Till** | $\tau_*$ at $N = 50$ (kPa) | $\varphi$ | $c_o$ (kPa) | **Method** |
|---|---|---|---|---|
| Whillans Ice Stream | 2–4 | | | lab |
| Storglaciären | 29 | 26° | 5 | lab |
| Black Rapids | 43 | 40° | 1.3 | lab |
| Breidamerkur | 35 | 32° | 3.7 | in situ |
| Two Rivers (Laurentide) | 31 | 18° | 14 | lab |

## 4.3 Implications of Mohr–Coulomb Behavior

The Mohr–Coulomb description has several important consequences for glacier dynamics:

1. **Water pressure is the primary control.** Because $\tau_* \propto N$, the strength of the bed is set almost entirely by the effective pressure. To produce widespread till deformation beneath a glacier 100 m thick (overburden ∼900 kPa), the water pressure must exceed about 83% of overburden. Beneath 1 km of ice, the threshold is about 98% of overburden. The West Antarctic ice streams, where $P_w$ matches the 9 MPa overburden to within 1%, are the archetype.

2. **Plastic behavior means the drag is bounded.** A till that behaves plastically cannot support a shear stress greater than $\tau_*$. If the applied stress exceeds $\tau_*$, the till fails and the glacier accelerates. This is fundamentally different from a viscous bed, where the drag would increase with velocity.

3. **Dilatancy provides a negative feedback.** When a consolidated till begins to shear, it dilates (expands), creating new pore space. If water cannot flow in fast enough to fill this new space, the pore pressure drops and $N$ increases, strengthening the till. This provides a negative feedback that can stabilize deformation. If, on the other hand, water flows efficiently to the dilating zone, the pressure remains high and deformation continues.

4. **Deformation is episodic.** Subglacial observations show that till deformation varies on timescales of hours to days, is concentrated near the top of the till layer, and often occurs on discrete slip planes rather than as distributed shear. The rate and location of deformation are sensitive to water pressure fluctuations.

# 5 Sliding Relations for Glacier Models

Glacier flow models require a relation between basal velocity $u_b$, basal shear stress $\tau_b$, and other variables that serves as the basal boundary condition. Because no complete theory of basal sliding exists, such relations are necessarily phenomenological.

## 5.1 The Weertman Sliding Law

The simplest and most widely used sliding relation has the form

$$\boxed{u_b = C\,\tau_b^m,} \tag{21}$$

where $C$ is a sliding coefficient (inverse of the drag factor) and $m$ is the sliding exponent. Weertman's theory gives $m = (n+1)/2 \approx 2$ for $n = 3$. In practice, $m$ and $C$ are treated as tuning parameters. This law does not include the effect of water pressure and is appropriate only in the local-control regime.

## 5.2 Effective-Pressure-Dependent Sliding Laws

To include the effect of water pressure, the Weertman law is generalized to

$$u_b = k\,\frac{\tau_b^p}{N^q}, \tag{22}$$

with $p$ and $q$ positive exponents and $k$ a coefficient that depends on bed properties. For the simple case $p = 3$ and $q = 1$, this is equivalent to

$$u_b = k\,\tau_b^3\,N^{-1}. \tag{23}$$

This form captures the first-order observation that sliding velocity increases with basal shear stress and decreases with effective pressure. It was used by [1] and tested against observations of Variegated Glacier following a surge, though no single pair $(p, q)$ could explain velocity variations over both time and space.

In glacier flow models, the Weertman law (equation 21) is often written in its inverse form (basal stress as a function of velocity):

$$\tau_b = C^{-1/m}\, u_b^{1/m}, \tag{24}$$

where the sliding coefficient $C$ absorbs all bed properties including roughness, water pressure, and till strength. This is the form used in the SSA equations studied in the dynamics lectures.

### 5.3 Form Drag and Skin Friction: Two Categories of Resistance

Before presenting specific sliding laws, it is useful to distinguish the two fundamental mechanisms by which a glacier bed resists the overlying flow [12]:

- **Form drag** arises from the pressure gradient associated with flow around bed obstacles. Ice must deform to pass over and around bumps, and the asymmetry of normal stress (high on the upstream side, low on the downstream side) produces a net drag in the flow direction. Form drag depends strongly on slip rate (faster flow means more intense deformation around obstacles) but does not depend directly on effective pressure. It is the dominant resistance mechanism on a rigid bed when the ice is in full contact with the bed (no cavities).
- **Skin friction** arises from frictional resistance at the ice–bed contact. On a hard bed, skin friction comes from rock–rock contact between debris-laden basal ice and the bedrock; on a deformable bed, it comes from the yield strength of the till at grain-to-grain contacts. Skin friction depends strongly on effective pressure $N$ (through the Mohr–Coulomb criterion, $\tau_* = f\,N$) but is essentially independent of slip rate—it is a rate-independent, plastic process.

The total basal drag is the sum of form drag and skin friction (Figure 6). Cavity formation mediates the transition between the two regimes: at low slip rates, the ice is in full contact with the bed and form drag dominates; at high slip rates, extensive cavitation reduces the ice–bed contact area, diminishing form drag and leaving skin friction as the primary resistance mechanism. In the Coulomb limit ($u_b \to \infty$ or $N \to 0$), only skin friction remains and $\tau_b \to f\,N$.

### 5.4 The Regularized Coulomb Sliding Law

The physical picture of form drag transitioning to skin friction motivates a sliding law that smoothly interpolates between the two regimes. [15] derived such a law from the Nye–Kamb sliding problem with cavitation (equation 19). [5] obtained a similar form from finite-element simulations. The general structure is:

$$\boxed{\tau_b = N\, f(\Lambda), \quad \Lambda = \frac{u_b}{N^n},} \tag{25}$$

where $f(\Lambda)$ is a function that increases monotonically from zero, reaches a maximum, and then remains bounded. A widely used parameterization [15, 5] is

$$\tau_b = C\,N \left( \frac{\Lambda}{\Lambda + \Lambda_0} \right)^{1/n}, \tag{26}$$

where $C$ is a constant less than $\tan\beta$ (the maximum bed slope) and $\Lambda_0$ is a reference velocity scale. At low velocities ($\Lambda \ll \Lambda_0$), $\tau_b \propto u_b^{1/n}$—the form-drag (Weertman) regime. At high velocities ($\Lambda \gg \Lambda_0$), $\tau_b \to C\,N$—the skin-friction (Coulomb) regime.

This sliding law has two physically desirable features:

- It transitions smoothly between rate-strengthening behavior at low velocities (form drag dominates) and rate-independent behavior at high velocities (skin friction dominates).
- It ensures that drag cannot exceed the Iken bound or the Mohr–Coulomb yield stress, providing a physical upper limit on basal resistance.

## 5.5 Experimental Confirmation: Zoet & Iverson (2020)

[20] provided the first experimental evidence that the regularized Coulomb form applies to deformable beds. Using a ring-shear device, they slid pressurized ice at its melting temperature over a water-saturated till bed and measured the steady-state relationship between slip resistance and slip velocity. Two configurations were tested: till without clasts (smooth bed) and till with centimeter-scale clasts embedded in the surface (rough bed).

In both experiments, the results showed two distinct regimes:

1. At low slip velocities, drag increased with velocity (rate-strengthening). This is the form-drag regime: ice deforms viscously around static clasts, and faster sliding produces more drag.
2. Above a transition velocity $u_t$, drag became independent of velocity and equal to the Coulomb shear strength of the till, $\tau_* = N\,\tan\varphi$. This is the skin-friction regime: clasts are mobilized by the sliding ice and begin to plow through the yielding till.

The transition between the two regimes occurs when the drag exerted by ice flow around a clast equals the force required to plow the clast through the till at its Coulomb strength. For clasts with larger radii, the transition velocity is lower because the clast protrudes further into the till and engages more material. The transition velocity $u_t$ depends on the effective viscosity of the ice, the clast size, the effective pressure, and the till strength.

[20] proposed a slip law that captures both regimes:

$$\boxed{\tau_b = N\,\tan\varphi\left(\frac{u_b}{u_b+u_t}\right)^{1/p},} \tag{27}$$

where $\varphi$ is the friction angle of the till and $p$ is an empirical exponent ($p \approx 5$ in their experiments). This has the same mathematical form as equation (26), with $C = \tan\varphi$, $\Lambda_0 = u_t/N^n$, and $1/p$ playing the role of $1/n$. The fact that the same functional form emerges for both hard beds (from the Schoof/Gagliardini cavity theory) and soft beds (from the Zoet & Iverson experiments) is a strong indication that the regularized Coulomb law captures the essential physics of basal slip.

## 5.6 Toward a Universal Slip Law

[12] argued that the similarity between the hard-bed and soft-bed sliding laws is not a coincidence but reflects the shared underlying physics: total drag is the sum of form drag and skin friction, and the transition between the two is governed by the same competition regardless of bed type.

**Key point: the case for a universal slip law**

On a hard bed, form drag arises from viscous flow around bedrock bumps. Cavities form downstream of bumps when $P_w$ exceeds the separation pressure, reducing the ice–bed contact area and transitioning the system toward skin friction (debris–bedrock friction at the remaining contacts).

On a soft bed, form drag arises from viscous flow around static clasts embedded in the till surface. When the drag on a clast exceeds the yield strength of the surrounding till, the clast is mobilized and plows through the till. The resistance is then set by the Coulomb strength of the till, independent of velocity.

In both cases, the total drag takes the form

$$\tau_b = \min\Big[C_{\text{form}}\, u_b^{1/m}, \;\; f\, N\Big], \tag{28}$$

regularized to produce a smooth transition. The regularized Coulomb law (equation 26 or 27) is the simplest parameterization that captures this behavior. The same functional form—rate-strengthening at low velocity, rate-independent at high velocity—applies irrespective of bed type, with different physical interpretations of the parameters $C$, $u_t$ (or $\Lambda_0$), and $f$.

### 5.7 Which Law to Use?

The Weertman law ($u_b = C\,\tau_b^m$) remains the most common choice in ice-sheet models because of its simplicity and because $C$ can be tuned spatially. However, it allows unphysically large drag at high velocities and lacks effective-pressure dependence. The Budd law ($u_b = k\,\tau_b^p/N^q$) adds $N$-dependence but still permits unbounded drag. Figure 7 compares the three laws at fixed $N$.

The regularized Coulomb law (equations 26 and 27) is the physically preferred form for three reasons:

1. It respects Iken's bound and Mohr–Coulomb behavior.
2. It is derived from first principles for hard beds [15] and confirmed experimentally for soft beds [20].
3. It has the same functional form regardless of bed type, requiring only reinterpretation of the parameters—making it applicable without prior knowledge of whether the bed is hard or soft [12].

For marine ice sheets near grounding lines, where $N$ is small and the Coulomb regime dominates, the regularized Coulomb law produces qualitatively different dynamics than the Weertman law—including different grounding-line flux sensitivities and stability properties.

## 6 Worked Example: Effective Pressure and Sliding Speed

Consider a glacier 500 m thick resting on hard bedrock with roughness $R = 0.1$. The bed has sinusoidal undulations with $\lambda = 10$ m and $a = 1$ m, so $a/\lambda = R$. The basal shear stress is $\tau_b = 100$ kPa. We compute the separation pressure, the Iken bound, and the sliding velocity for two values of water pressure.

**Step 1: Ice-overburden and separation pressures.**

$$P_i = \rho_i\, g\, H = 917 \times 9.8 \times 500 = 4.49 \text{ MPa}, \tag{29}$$

$$P_s = P_i - \frac{\lambda\,\tau_b}{a\,\pi} = 4.49 - \frac{10 \times 0.1}{1 \times \pi} = 4.49 - 0.318 = 4.17 \text{ MPa}. \tag{30}$$

The water pressure must exceed 4.17 MPa (93% of overburden) for cavities to form.

**Step 2: Iken's bound.** The maximum bed slope is $\tan\beta = 2\pi a/\lambda = 2\pi \times 0.1 = 0.628$, so:

$$\tau_b^{\max} = N\,\tan\beta = N \times 0.628. \tag{31}$$

For the applied $\tau_b = 100$ kPa, sliding becomes unstable when

$$N < \frac{\tau_b}{\tan\beta} = \frac{100}{0.628} = 159 \text{ kPa}, \quad \text{i.e., } P_w > 4.49 - 0.159 = 4.33 \text{ MPa}. \tag{32}$$

At water pressures above 96% of overburden, the bed can no longer support the applied basal drag.

**Step 3: Sliding velocities.** Using the effective-pressure-dependent law $u_b = k\,\tau_b^3\,N^{-1}$ with $k = 3.5 \times 10^{-16}\ \mathrm{m\,Pa^{-2}\,s^{-1}}$ (a representative value):

| $P_w$ (MPa) | $P_w/P_i$ | $N$ (MPa) | $u_b$ ($\mathrm{m\,yr^{-1}}$) |
|---|---|---|---|
| 3.59 | 80% | 0.90 | 12 |
| 4.26 | 95% | 0.22 | 50 |
| 4.40 | 98% | 0.09 | 123 |

Increasing the water pressure from 80% to 98% of overburden increases the sliding speed by an order of magnitude (Figure 8). This extreme sensitivity to water pressure is why the subglacial hydrological system—which controls the distribution and magnitude of $P_w$—is a first-order control on glacier dynamics.

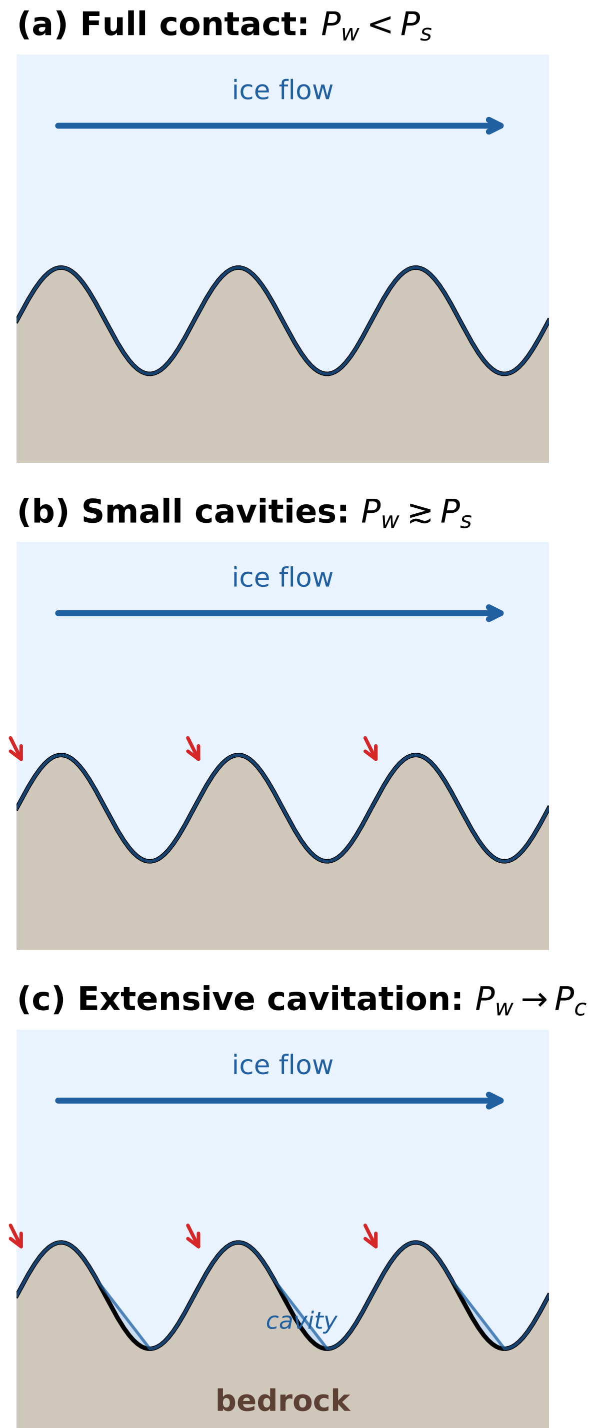


Figure 3: Progressive cavity formation on a sinusoidal bed as water pressure increases. **(a)** Full ice–bed contact when $P_w < P_s$. **(b)** Small water-filled cavities form in the lee of bumps once $P_w$ exceeds the separation pressure $P_s$. Red arrows indicate stress concentrations on the remaining upstream contact areas. **(c)** Extensive cavitation as $P_w \to P_c$; ice contacts only the upstream faces of bumps, and the effective roughness is greatly reduced.

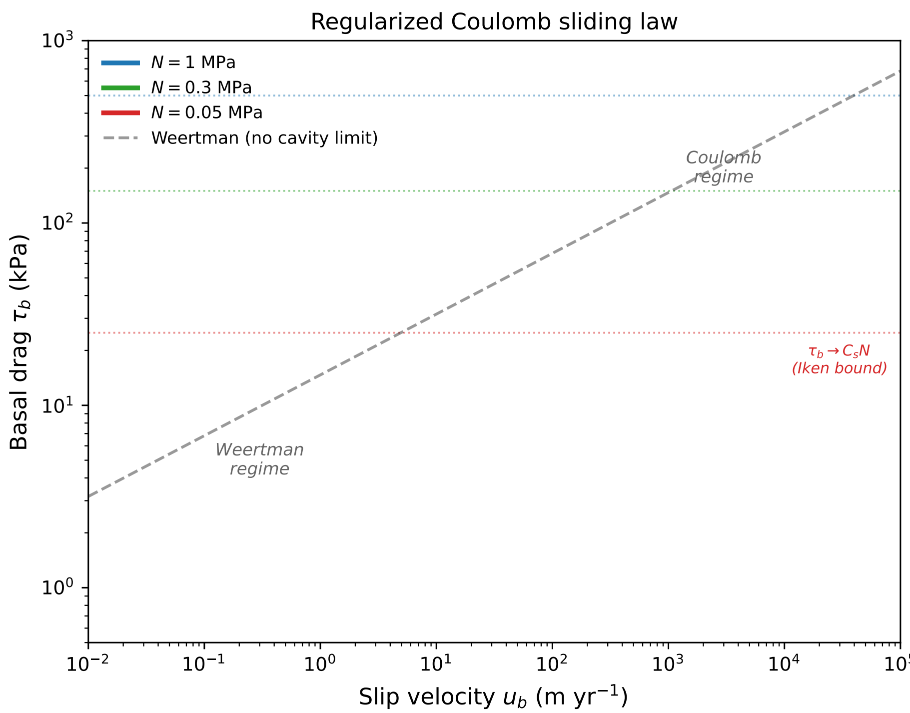


Figure 4: Basal drag $\tau_b$ vs. slip velocity $u_b$ for the regularized Coulomb sliding law (equation 19) at three values of effective pressure $N$. At low velocities, drag increases following Weertman-type scaling ($\tau_b \propto u_b^{1/n}$). At high velocities, drag saturates at the Iken/Coulomb bound $\tau_b \rightarrow C_s N$ (dotted lines), where $C_s = \tan\beta$ is the maximum bed slope. Lower effective pressure reduces both the drag maximum and the Coulomb limit.

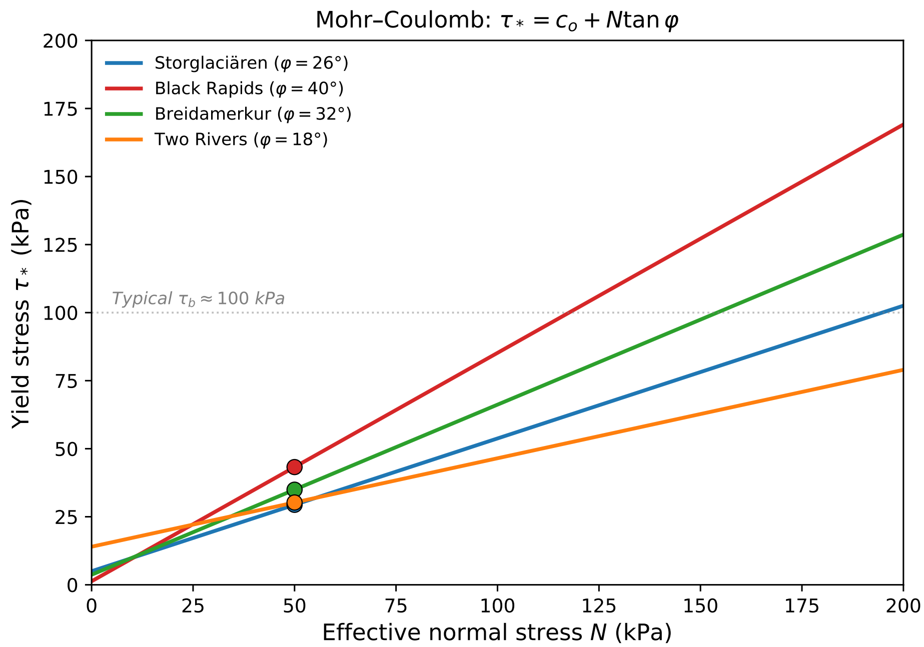


Figure 5: Mohr–Coulomb yield stress $\tau_* = c_o + N \tan\varphi$ as a function of effective normal stress $N$ for four subglacial tills with measured friction angles. Circles mark the yield stress at $N = 50$ kPa. The horizontal line at $\tau_b = 100$ kPa indicates a typical basal shear stress; tills below this line are too weak to support the applied stress at the corresponding $N$, and the bed will deform. Data from [2], Table 7.5.

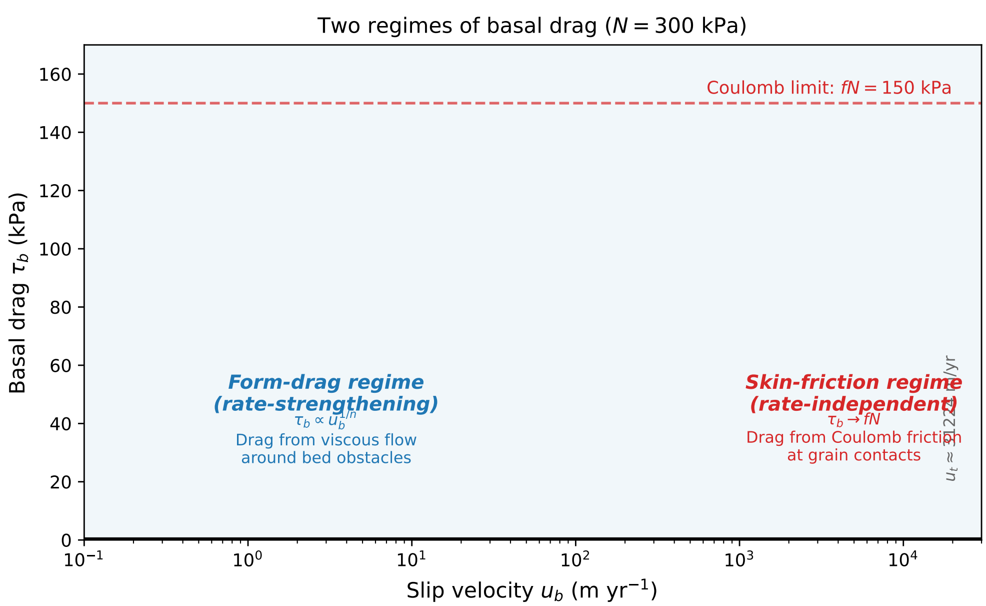


Figure 6: The two regimes of basal drag illustrated for the regularized Coulomb sliding law at $N = 300$ kPa. At low slip velocities (blue shading), form drag dominates: drag increases with velocity as ice deforms around bed obstacles. Above the transition velocity $u_t$, skin friction dominates (red shading): drag saturates at the Coulomb limit $f\,N$, independent of velocity. The transition velocity depends on bed properties, ice viscosity, and effective pressure.

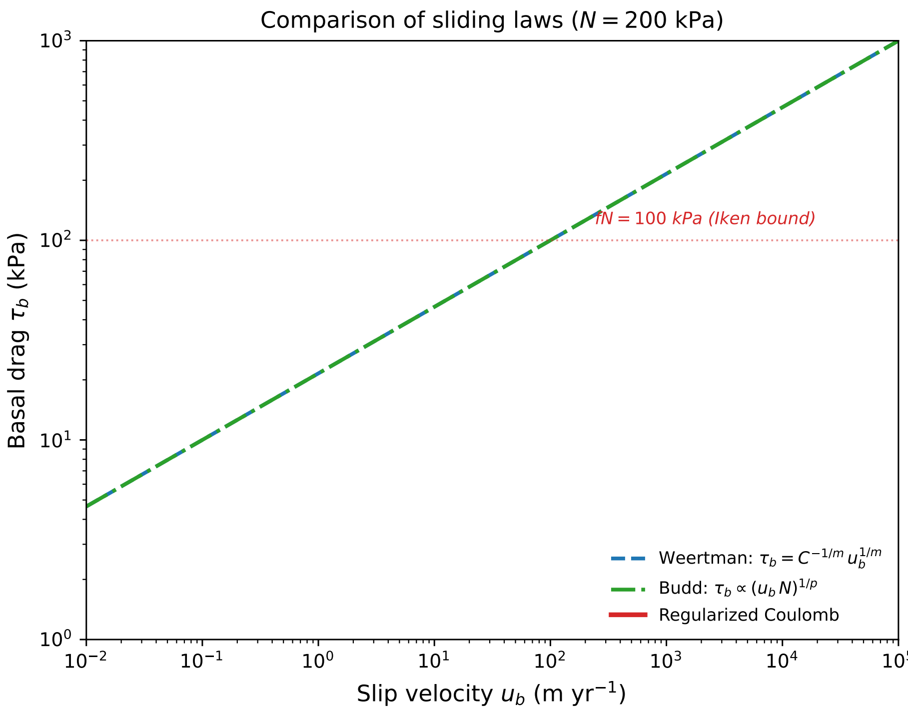


Figure 7: Comparison of three sliding laws at fixed effective pressure $N = 200$ kPa. The Weertman law (dashed) and Budd law (dash-dot) both allow drag to grow without bound as velocity increases. The regularized Coulomb law (solid) transitions from Weertman-like behavior at low velocities to a Coulomb limit $\tau_b \to f\,N$ at high velocities, consistent with Iken's bound, Mohr–Coulomb till behavior, and the experimental results of [20].

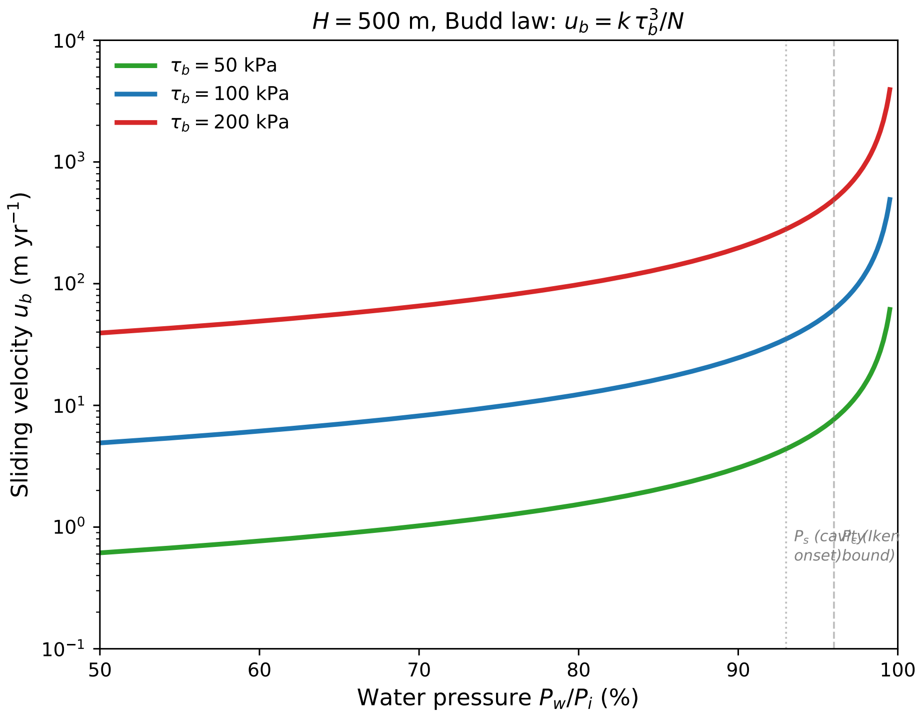


Figure 8: Sliding velocity as a function of water pressure (expressed as a fraction of ice-overburden pressure) for the Budd-type sliding law $u_b = k\,\tau_b^3/N$ at three values of basal shear stress. The vertical dotted lines mark the approximate separation pressure $P_s$ (onset of cavitation) and the critical pressure $P_c$ (Iken bound) for the bed geometry of the worked example. Sliding speed increases by more than an order of magnitude as water pressure rises from 80% to 98% of overburden.

## Reading

Cuffey & Paterson, *Physics of Glaciers*, 4th ed.:

- Section 7.1: Introduction and measurements of basal velocity
- Section 7.2: Hard beds—Weertman's theory, cavities, water pressure, and Iken's bound
- Section 7.3: Deformable beds—till properties and Mohr–Coulomb behavior

# A Ice Stream Stagnation: The Undrained Plastic Bed Model

West Antarctic ice streams flow at hundreds of meters per year over beds of weak, water-saturated till with basal shear stresses of only a few kilopascals. Despite this apparent simplicity, individual ice streams can stagnate on timescales of centuries—the best-documented example being Kamb Ice Stream (formerly Ice Stream C), which shut down approximately 150 years ago. [18] proposed a thermomechanical model—the **undrained plastic bed (UPB) model**—that explains how small perturbations in the basal energy balance can trigger transitions between fast-flowing and stagnant states. This appendix summarizes the key elements of the model.

## A.1 The Compressible-Coulomb-Plastic Till Model

The UPB model is built on the laboratory results of [17], who showed that the till beneath Ice Stream B (now Whillans Ice Stream) behaves as a Coulomb-plastic material with three interrelated state variables: shear strength $\tau_f$, effective normal stress $N$, and porosity $\phi_p$ (the fraction of total volume occupied by pore water).

The till strength follows the Mohr–Coulomb criterion with negligible cohesion:

$$\tau_f = N\ \tan\varphi, \qquad \varphi \approx 24^\circ. \tag{33}$$

The porosity depends logarithmically on effective stress. [17] express this in terms of the void ratio $e = \phi_p/(1-\phi_p)$:

$$e = e_0 - C_c\ \log\left(\frac{N}{N_0}\right), \tag{34}$$

where $e_0$ is a reference void ratio at $N_0 = 1$ kPa and $C_c \approx 0.12$ is the dimensionless coefficient of compressibility. Together, equations (33) and (34) can be combined to express till strength as a function of porosity:

$$\tau_f = a\ \exp\left(\frac{-b\,\phi_p}{1-\phi_p}\right), \qquad a = 944{,}000\ \text{kPa}, \quad b = 21.7, \tag{35}$$

which shows that till strength is exponentially sensitive to porosity. A change in porosity from 0.35 to 0.39 (about 4 percentage points) changes the till strength by nearly an order of magnitude. Figure 9 illustrates this relationship.

## A.2 Feedback Between Water Storage, Till Strength, and Basal Melting

The key physical insight of the UPB model is that the rate of water storage in till pore spaces couples till strength to the basal thermal energy balance. The rate of change of porosity is

$$\frac{\partial \phi_p}{\partial t} = (1-\phi_p)^2\,\frac{m_r - d_r}{Z_s}, \tag{36}$$

where $m_r$ is the basal melt rate, $d_r$ is the drainage rate, and $Z_s$ is the thickness of the active till layer. (The factor $(1-\phi_p)^2$ arises from converting the void ratio rate $\partial e/\partial t = (m_r - d_r)/Z_s$ used by [18] to porosity via $e = \phi_p/(1-\phi_p)$.)

The basal melt rate is determined by the thermal energy balance at the ice–bed interface:

$$\boxed{m_r = \frac{\tau_b\,U_b + G - k_i\,\Theta_b}{L_i\,\rho_{\text{ice}}},} \tag{37}$$

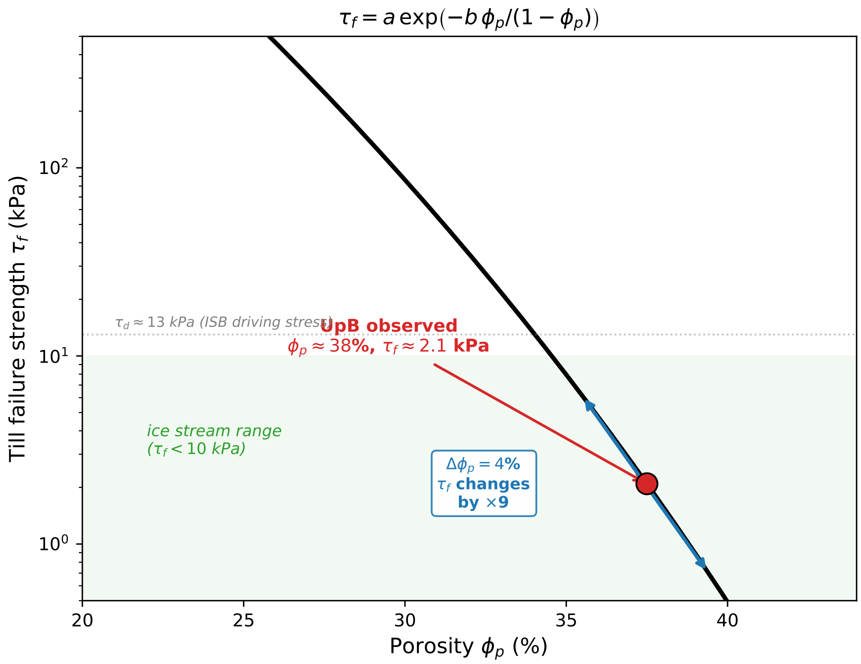


Figure 9: Till failure strength $\tau_f$ as a function of porosity $\phi_p$ for the UpB till, following the empirical fit of [17]. The red circle marks the observed conditions beneath Ice Stream B ($\phi_p \approx 0.38$, $\tau_f \approx 2$ kPa). A change in porosity of only 4 percentage points (blue arrows) changes the till strength by nearly an order of magnitude. The green shading marks the strength range relevant to ice streaming ($\tau_f < 10$ kPa).

where $\tau_b\,U_b$ is the shear heating rate, $G$ is the geothermal heat flux, $k_i\,\Theta_b$ is the conductive heat loss (with $\Theta_b$ the basal temperature gradient), and $L_i\,\rho_{\text{ice}}$ converts energy flux to melt rate. If $m_r > 0$, the base is melting; if $m_r < 0$, the base is freezing.

The coupling works as follows. Because the till is a Coulomb-plastic material, the basal shear stress equals the till strength: $\tau_b = \tau_f$. The till strength depends on porosity through equation (35), and the porosity changes in response to the melt rate through equation (36). The melt rate in turn depends on the shear heating $\tau_b\,U_b$, which depends on the till strength. This closes the loop: a small change in till water content affects its strength, which affects the shear heating, which affects the melt rate, which further changes the water content.

### A.3 Ice Stream Velocity on a Plastic Bed

To complete the model, [18] derived the relationship between ice stream velocity and till strength for a rectangular channel. For an ice stream of half-width $W$ and driving stress $\tau_d$ flowing over a bed with uniform strength $\tau_b$, the centerline velocity is

$$U_b = \left(\frac{\tau_d - \tau_b}{\tau_d}\right)^n W^{n+1}\,U_d = \left(1 - \frac{\tau_b}{\tau_d}\right)^n W^{n+1}\,U_d, \tag{38}$$

where $U_d = 2^{1-n}\,\tau_d^n\,(n+1)^{-1}\,B_n^{-1}$ is the velocity that would result from internal deformation alone, and $n$ and $B_n$ are Glen's flow law parameters. The marginal shear stress $(\tau_d - \tau_b)$ is the fraction of the driving stress not supported by the bed, which must be borne by the ice stream margins.

Equation (38) predicts that the maximum ice stream velocity occurs when the bed strength is zero ($\tau_b = 0$); when $\tau_b = \tau_d$, the bed supports the entire driving stress and the sliding velocity is zero.

## A.4 Two Stable Modes and the Condition for Stagnation

Substituting $\tau_b = \tau_f$ into equation (37) and using equation (38) for $U_b$ yields the melt rate as a function of till strength alone (for fixed driving stress and geometry). [18] showed that the resulting function $m_r(\tau_f)$ has a maximum at $\tau_b = (n+1)^{-1}\,\tau_d = 0.25\,\tau_d$ for $n = 3$.

At steady state, the melt rate must equal zero (in the undrained case, $d_r = 0$) or equal the drainage rate. The condition $m_r = 0$ defines two equilibrium states (Figure 10):

1. **Ice stream mode** (stable): $\tau_b < 0.25\,\tau_d$. The bed is weak, the ice stream is fast, and shear heating is substantial. If a perturbation weakens the bed slightly ($\tau_b$ decreases), the shear heating $\tau_b\,U_b$ also decreases (because we are on the left side of the maximum), reducing the melt rate. This causes freeze-on, which strengthens the till, restoring the system toward equilibrium. This is a negative feedback.

2. **Unstable equilibrium**: $\tau_b > 0.25\,\tau_d$. The second zero crossing of $m_r$ (open circle in Figure 10) is unstable. If a perturbation weakens the bed, shear heating increases (because we are on the right side of the maximum, where $\partial(\tau_b U_b)/\partial\tau_b > 0$), which increases melting, further weakening the bed—a positive feedback that pushes the system toward the ice stream mode. If instead the bed strengthens, shear heating decreases, leading to freeze-on that further strengthens the bed, pushing the system toward the **ice sheet mode**—a stagnant state in which the base is frozen and $\tau_b \approx \tau_d$.

The boundary between the two modes is the saddle-node bifurcation at $\tau_f = (n+1)^{-1}\,\tau_d$:

$$\boxed{\tau_f^{\mathrm{crit}} = \frac{\tau_d}{n+1}.} \tag{39}$$

For $n = 3$, ice streaming requires $\tau_f < 0.25\,\tau_d$. With typical driving stresses of 10–30 kPa for West Antarctic ice streams, this requires till strengths below 2.5–7.5 kPa—consistent with the $\sim$2 kPa measured beneath Whillans Ice Stream.

## A.5 Implications

The UPB model produces several results that bear directly on ice sheet dynamics and sea-level projections:

1. **Ice stream stagnation is thermally triggered.** A small increase in conductive heat loss (e.g., from thinning of the ice column) or decrease in geothermal flux can shift the $m_r = 0$ curve below zero for all bed strengths in the ice stream mode, eliminating the stable equilibrium and forcing the system to the ice sheet mode. The transition can be rapid—the characteristic timescale for till consolidation driven by basal freeze-on is of order 10 years.

2. **Till water content is a first-order state variable.** The exponential sensitivity of till strength to porosity (equation 35) means that changes in water storage of a few percentage points can produce order-of-magnitude changes in basal resistance. Models that treat till strength as a prescribed parameter miss this feedback.

3. **The model is consistent with observations from Kamb Ice Stream.** Kamb Ice Stream stagnated ∼150 years ago despite having a driving stress (∼13 kPa) similar to neighboring active ice streams. The UPB model predicts that a slight loss of basal heat (from margin migration or changes in ice geometry) could have tipped the system from the ice stream mode to the ice sheet mode.

4. **The ice shelf mode.** If the geothermal flux exceeds the conductive heat loss ($G > k_i\,\Theta_b$), a third stable mode exists in which the till is extremely weak ($\tau_f \to 0$, $\phi_p$ approaches unity) and the ice stream velocity approaches the ice-shelf limit $U_b(\tau_b = 0) = \tau_d^n\, W^{n+1}\, U_d$.

The UPB model demonstrates that the stability of ice streams is governed by a subtle interplay between basal thermodynamics, till mechanics, and ice flow—and that the same physics that enables fast ice-stream flow can also bring it to an abrupt halt.

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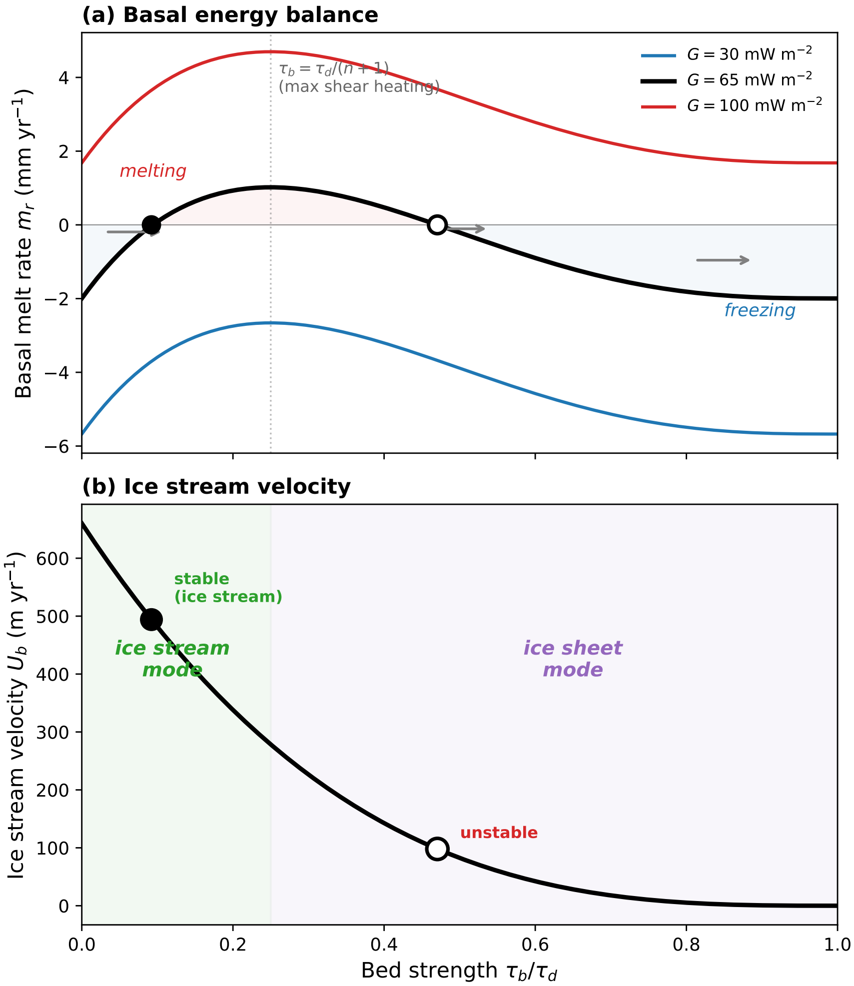


Figure 10: Stability of the undrained plastic bed (UPB) model, computed for Ice Stream B parameters ($\tau_d = 13$ kPa, $\Theta_b = 0.04$ K m$^{-1}$). **(a)** Basal melt rate as a function of bed strength for three geothermal heat flux values. The maximum shear heating occurs at $\tau_b = \tau_d/(n+1) = 0.25\,\tau_d$. At $G = 65$ mW m$^{-2}$ (black), two equilibria exist where $m_r = 0$: the ice stream mode (stable, filled circle) at low $\tau_b$ and the ice sheet mode (unstable, open circle) at high $\tau_b$. At $G = 30$ mW m$^{-2}$ (blue), the melt rate is negative for all bed strengths—the base freezes everywhere and no ice-stream equilibrium exists. At $G = 100$ mW m$^{-2}$ (red), the melt rate is positive everywhere and the bed weakens toward the ice-shelf limit. Arrows show the direction of evolution for the reference case: melting weakens the bed (leftward) and freezing strengthens it (rightward). **(b)** Corresponding ice stream velocity, calibrated to Whillans Ice Stream observations ($U_b \approx 400$ m yr$^{-1}$ at $\tau_b \approx 2$ kPa). After [18].